CS 506: Quantum Computing

Scribe Notes: Period Recovery via QFT and Continued Fractions

Kanglin Xu

April 13, 2026

1 Overview

This lecture completes the quantum-to-classical pipeline in Shor's algorithm. We first analyze the QFT measurement distribution and establish that the measurement outcome y_m approximates a rational multiple of 2^m with denominator r, with constant total probability. We then develop Euclid's algorithm and continued fractions as the classical tools that recover r from y_m.

2 Setup

Let $N = 2^n$ be the integer to factor, with $n = \log_2 N$. We choose $M = 2^m$ as the smallest power of two satisfying

$$2N^2 \;\le\; M = 2^m \;<\; 4N^2, \qquad \text{so} \quad m = \lceil \log_2(2N^2) \rceil \;\le\; 2\log_2 N + 2.$$

Let Q denote the number of terms in the periodic superposition; we have $Q \approx 2^m/r$, with

$$\left\lfloor \frac{2^m}{r} \right\rfloor \;\ge\; 1 \qquad \text{and} \qquad \frac{2^m}{4r} \;\ge\; 1.$$

3 Post-Measurement State and QFT

After applying Hadamard gates, the oracle U_f, and measuring the second register, the first register collapses to

$$|\phi\rangle \;=\; \frac{1}{\sqrt{Q}} \sum_{z=0}^{Q-1} |x_0 + zr\rangle,$$

where r is the unknown period and x_0 is the observed offset. Applying the m-qubit QFT gives the state

$$|\psi\rangle \;=\; \mathrm{QFT}_{2^m}|\phi\rangle,$$

which we then measure to obtain $y_m \in \{0, 1, \ldots, 2^m - 1\}$.

4 Measurement Probability Analysis

4.1 Exact Formula

For each $k \in \{0, 1, \ldots, r-1\}$ write the candidate outcome as

$$y_m \;=\; k \cdot \frac{2^m}{r} + \delta_k, \qquad |\delta_k| \;\le\; \frac{1}{2}.$$

The factored-out global phase $e^{2\pi i x_0 y_m/2^m}$ has modulus 1 and drops out, giving

$$\Pr\left[y_m = k \cdot \frac{2^m}{r} + \delta_k\right] \;=\; \frac{1}{2^m Q} \cdot \frac{\sin^2(Q\pi r\delta_k/2^m)}{\sin^2(\pi r\delta_k/2^m)}.$$

4.2 Lower Bound for Fixed k

We apply two standard estimates. Set $\alpha = \pi r\delta_k/2^m$ (the denominator angle). Since $Q \approx 2^m/r$, the numerator angle is $Q\alpha \approx \pi\delta_k$, which satisfies $|Q\alpha| \le \pi/2$ because $|\delta_k| \le 1/2$.

- **Numerator lower bound.** For $|x| \le \pi/2$, $|\sin x| \ge (2/\pi)|x|$, so

$$\sin^2(Q\alpha) \;\ge\; \frac{4}{\pi^2}(Q\alpha)^2.$$

- **Denominator upper bound.** $\sin^2(\alpha) \le \alpha^2$.

Combining and substituting $Q = 2^m/r$:

$$\Pr\left[y_m = k \cdot \frac{2^m}{r} + \delta_k\right] \;\ge\; \frac{1}{2^m Q} \cdot \frac{(4/\pi^2)(Q\alpha)^2}{\alpha^2} \;=\; \frac{4Q}{\pi^2 \cdot 2^m} \;\approx\; \frac{4}{\pi^2 r} \;\ge\; \frac{0.4}{r},$$

since $4/\pi^2 \approx 0.405 \ge 0.4$.

4.3 Total Success Probability

The r values of $k \in \{0, 1, \ldots, r-1\}$ yield disjoint measurement outcomes, so the probabilities sum:

$$\Pr\left[y_m = k \cdot \frac{2^m}{r} + \delta_k \text{ for some } k\right] \;\ge\; r \cdot \frac{0.4}{r} \;=\; 0.4.$$

4.4 The Key Rational Approximation

From $y_m = k \cdot (2^m/r) + \delta_k$ we obtain

$$\frac{y_m}{2^m} \;=\; \frac{k}{r} + \frac{\delta_k}{2^m}.$$

Using $|\delta_k| \le 1/2$ and $2^m \ge 2N^2 \ge 2r^2$ (since $r \le N$):

$$\left|\frac{y_m}{2^m} - \frac{k}{r}\right| \;=\; \frac{|\delta_k|}{2^m} \;\le\; \frac{1}{2 \cdot 2^m} \;\le\; \frac{1}{4r^2}. \qquad (*)$$

The remaining challenge is classical: given the observed integer y_m, recover the fraction k/r (and hence r) from the approximation $y_m/2^m$.

5 Euclid's Algorithm

Euclid's algorithm (ca. 300 BC) computes $\gcd(x, y)$ for $x \le y$ by repeatedly replacing (x, y) with $(y \bmod x,\ x)$ until the remainder is zero, using the identity $\gcd(x, y) = \gcd(y \bmod x,\ x)$.

Example 1. Compute $\gcd(189,\ 263)$ with $x = 189$, $y = 263$.

$$\begin{aligned}
263 &= 1 \cdot 189 + 74, & d_1 &= 1,\ r_1 = 74 \\
189 &= 2 \cdot 74 + 41, & d_2 &= 2,\ r_2 = 41 \\
74 &= 1 \cdot 41 + 33, & d_3 &= 1,\ r_3 = 33 \\
41 &= 1 \cdot 33 + 8, & d_4 &= 1,\ r_4 = 8 \\
33 &= 4 \cdot 8 + 1, & d_5 &= 4,\ r_5 = 1 \\
8 &= 8 \cdot 1 + 0. & d_6 &= 8,\ r_6 = 0
\end{aligned}$$

The last nonzero remainder is 1, so $\gcd(189,\ 263) = 1$.

6 Polynomial Runtime

Claim 1 (Without proof)**.** *After every two consecutive steps of Euclid's algorithm, the larger of the two current numbers decreases by a factor of at least* 2*.*

This implies immediately that

$$\#\text{ of steps } \leq\ 2\log_2 N,$$

which is polynomial in the input size $\log N$.

7 Continued Fraction Expansions

Definition 1. The **continued fraction expansion** of a positive rational b is

$$b \;=\; a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{a_\ell}}}},$$

written compactly as $b = [a_0;\, a_1, a_2, \ldots, a_\ell]$, where $a_0 \geq 0$ and $a_i \geq 1$ for $i \geq 1$. The integers a_i are the *partial quotients*, and they coincide exactly with the successive division quotients d_i in Euclid's algorithm on (x, y).

Example 2. Unrolling the division steps for 263/189 from the example above:

$$\begin{aligned}
\frac{263}{189} &= 1 + \frac{74}{189} = 1 + \frac{1}{\frac{189}{74}} = 1 + \cfrac{1}{2 + \cfrac{41}{74}} \\
&= 1 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{33}{41}}} = 1 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{8}{33}}}} \\
&= 1 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{4 + \cfrac{1}{8}}}}}.
\end{aligned}$$

Thus $263/189 = [1;\ 2, 1, 1, 4, 8]$.

8 Convergents

Definition 2. The **j-th convergent** of $b = [a_0;\ a_1, \ldots, a_\ell]$ is the rational number p_j/q_j obtained by truncating the expansion after a_{j-1}. Each convergent is the best rational approximation to b among all fractions with denominator at most q_j.

For $263/189 = [1;\ 2, 1, 1, 4, 8]$ the first four convergents are:

$$\begin{aligned}
&\text{1st:} && 1 && = 1.000 \\
&\text{2nd:} && 1 + \frac{1}{2} = \frac{3}{2} && \approx 1.500 \\
&\text{3rd:} && 1 + \cfrac{1}{2 + \cfrac{1}{1}} = 1 + \frac{1}{3} = \frac{4}{3} && \approx 1.333 \\
&\text{4th:} && 1 + \cfrac{1}{2 + \cfrac{1}{1 + \cfrac{1}{1}}} = 1 + \frac{1}{\frac{5}{2}} = \frac{7}{5} && = 1.400
\end{aligned}$$

9 Hardy–Wright Theorem

The following classical result (from Hardy and Wright's *An Introduction to the Theory of Numbers*) is the key bridge between rational approximation and continued fractions.

Theorem 1 (Hardy–Wright)**.** *Let k and r be positive integers and let b be a positive rational number. If*

$$\left|\frac{k}{r} - b\right| \leq \frac{1}{2r^2},$$

then k/r is a convergent in the continued fraction expansion of b.

10 Applying Hardy–Wright to Recover k/r

10.1 The Bound $(*)$ Revisited

From the key inequality $(*)$ derived in Section 4,

$$\left|\frac{y_m}{2^m} - \frac{k}{r}\right| \leq \frac{1}{4r^2} \leq \frac{1}{2r^2}.$$

The final inequality simply observes $4r^2 \geq 2r^2$. The hypothesis of the Hardy–Wright theorem is satisfied with $b = y_m/2^m$, so:

The fraction k/r is a convergent in the continued fraction expansion of $y_m/2^m$.

Remark 1. *Which* convergent? Without proof, it is a fact that there is *one and only one* convergent of $y_m/2^m$ that equals k/r, and this convergent can be identified algorithmically.

10.2 Complete Recovery Procedure

Given measurement outcome y_m:

1. Run Euclid's algorithm on $(y_m,\ 2^m)$ to obtain the continued fraction expansion of $y_m/2^m$. This takes $O(\log N)$ steps.
2. Enumerate the convergents p_0/q_0, $p_1/q_1, \ldots$ in order.
3. For each convergent p_j/q_j, test whether $a^{q_j} \equiv 1 \pmod{N}$.
4. Return the smallest q_j passing the test as the recovered period r.

Since Euclid's algorithm produces at most $O(\log N)$ convergents, the entire classical post-processing runs in polynomial time.

11 Key Takeaways

- With probability ≥ 0.4, the QFT measurement outcome y_m satisfies $|y_m/2^m - k/r| \leq 1/(4r^2)$ for some k. Per-candidate probability is $\geq 4/(\pi^2 r) \geq 0.4/r$.
- Euclid's algorithm computes $\gcd(x, y)$ in $\leq 2\log_2 N$ steps, and its successive quotients are the continued fraction coefficients of y/x.
- Convergents give the best rational approximations with bounded denominator.
- Hardy–Wright: any fraction within $1/(2r^2)$ of b must appear as a convergent of b.
- Combining $(*)$ with Hardy–Wright shows k/r is the *unique* recoverable convergent of $y_m/2^m$, completing the classical post-processing of Shor's algorithm.